

Key Concepts to Understand from Project 1 (Data Warehousing) and Project 2 (Data Mining)

Project 1: Data Warehousing (Fundamentals)

Core Concepts:

1. Data Warehouse Basics

- Definition: **Centralized repository** for integrated, historical, structured data.
- Purpose: Supports OLAP (analytical queries) rather than OLTP (transactions).

2. Star Schema Design

- **Fact Table:** Contains measurable metrics (e.g., sales revenue).
- **Dimension Tables:** Descriptive attributes (e.g., time, product, customer).
- **Entity-Relationship Diagram (ERD):** Visual representation of tables and relationships.

3. ETL Process (Extract, Transform, Load)

- **Extract:** Pull data from sources (e.g., CSV, APIs).
- **Transform:** Clean, standardize, and structure data (e.g., handling missing values).
- **Load:** Populate the PostgreSQL database.

4. OLAP Operations

- Aggregations (SUM, AVG, COUNT).
- Filtering (WHERE clauses).
- Joins (linking fact and dimension tables).

5. Tools & Implementation

- **PostgreSQL:** Database for storing the warehouse.
- **Python (Pandas):** ETL scripting.
- **SQL:** Querying and analysis.

Deliverables:

- Data Exploration Report.
- Star Schema ERD and Table Definitions.
- ETL Code Snippets. And - SQL Queries + Visualizations.

Project 2: Data Mining (Extension of Project 1)

Core Concepts:

1. Data Mining Basics

- Definition: Discovering patterns/insights from data.
- Relationship to Warehousing: Uses cleaned, structured data from the warehouse.

2. Key Data Mining Tasks

- **Association Rule Mining:** Find item relationships (e.g., "X $\rightarrow$ Y" purchases).
- **Classification:** Predict categorical outcomes (e.g., churn "yes/no").
- **Clustering:** Group similar data points (e.g., customer segments).
- **Regression:** Predict continuous values (e.g., sales forecasts).

3. Data Preparation for Mining

- **Feature Selection:** Choose relevant columns from the warehouse.
- **Data Transformation:** Scaling, encoding, splitting (train/test).

4. Algorithms & Tools

- **Apriori (Association Rules):** Via mlxtend.
- **K-Means (Clustering):** Via scikit-learn.
- **Logistic/Linear Regression:** Via scikit-learn.

5. Interpretation of Results

- Explain patterns (e.g., "Customers in Cluster 1 prefer Product A").
- Highlight actionable insights.

Deliverables:

- Short Report: Introduction, task relevance, code snippets, results.

Connection Between Projects

1. Data Flow:

- **Project 1 (Warehouse) → Structured Data → Project 2 (Mining).**

2. Complementary Goals:

- **Warehousing:** Organize data for analysis.
- **Mining:** Extract knowledge from organized data.

3. Tool Continuity:

- Both use Python, PostgreSQL, and Pandas.

Key Takeaways

- **Project 1:** Master **data modeling, ETL, and OLAP**.
- **Project 2:** Learn **mining techniques, preprocessing, and insight extraction**.

Data Set:

- World Bank Education Data: <https://data.worldbank.org/indicator/SE.ENR.PRIM.FM.ZS>
- dataset (API_SE.ENR.PRIM.FM.ZS) specifically tracks **gender parity in primary education enrollment** across countries and years.
- Where **Gender Parity Index (GPI)- F/M ratio** --- **<1, 1, >1**

API_SE.ENR.PRIM.FM.ZS_DS2_en_csv_v2_92648.csv - Microsoft Excel (Product Activation Failed)

FileHomeInsertPage LayoutFormulasDataReviewViewPDFelement

CutCopyPasteFormat PainterClipboardFontAlignmentNumberStylesCellsEditing

Calibri11GeneralConditional Formatting as Table Cell Insert Delete Format AutoSum Fill Sort & Find & Filter & Select

B I U Merge & Center \$ % , .00 %

A1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	Data Source	World Development Indicators																	
2																			
3	Last Update	#####																	
4																			
5	Country N	Country C	Indicator I	Indicator C	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
6	Aruba	ABW	School enr	SE.ENR.PRIM.FM.ZS															
7	Africa East	AFE	School enr	SE.ENR.PRIM.FM.ZS											0.75416	0.7559	0.76398	0.76969	0.78151
8	Afghanistan	AFG	School enr	SE.ENR.PRIM.FM.ZS											0.16919	0.16453	0.16825	0.17596	0.17467
9	Africa West	AFW	School enr	SE.ENR.PRIM.FM.ZS											0.62117	0.63836	0.65407	0.66763	0.67253
10	Angola	AGO	School enr	SE.ENR.PRIM.FM.ZS												0.62011	0.63192		
11	Albania	ALB	School enr	SE.ENR.PRIM.FM.ZS															
12	Andorra	AND	School enr	SE.ENR.PRIM.FM.ZS															
13	Arab World	ARB	School enr	SE.ENR.PRIM.FM.ZS											0.64789	0.64615	0.6517	0.65891	0.66558
14	United Arab Emirates	ARE	School enr	SE.ENR.PRIM.FM.ZS												0.68551	0.6648	0.73183	0.79601
15	Argentina	ARG	School enr	SE.ENR.PRIM.FM.ZS											1.00583	0.99948	0.99067	0.99154	0.98463
16	Armenia	ARM	School enr	SE.ENR.PRIM.FM.ZS															
17	American Samoa	ASM	School enr	SE.ENR.PRIM.FM.ZS															
18	Antigua and Barbuda	ATG	School enr	SE.ENR.PRIM.FM.ZS												0.9271			1.24355
19	Australia	AUS	School enr	SE.ENR.PRIM.FM.ZS												0.9965	0.998	0.99961	1.00063
20	Austria	AUT	School enr	SE.ENR.PRIM.FM.ZS												0.98864	0.9917	0.99448	0.99823
21	Azerbaijan	AZE	School enr	SE.ENR.PRIM.FM.ZS															
22	Burundi	BDI	School enr	SE.ENR.PRIM.FM.ZS												0.48402	0.49804	0.55856	0.58606
23	Belgium	BEL	School enr	SE.ENR.PRIM.FM.ZS												0.98892	0.98862	1.01055	0.99891
24	Benin	BDN	School enr	SE.ENR.PRIM.FM.ZS												0.44570	0.45472	0.45711	0.44845

Ready

Explanation of Data Mining Report (Project 2)

The report analyzes **global female-to-male enrollment ratios** using data mining techniques. Below is a step-by-step breakdown with clear explanations of key concepts:

1. Data Preparation

Goal: Clean and structure raw data for analysis.

Steps Taken:

- **Raw Data Format:**
 - Initially, data was in "long format" (columns: country_code, year, enrollment_ratio_fm).
 - **Transformed to "wide format" (pivot table):**
 - Each row = one country.
 - Each column = enrollment ratio for a specific year.
- **Handling Missing Values:**
 - Used **mean imputation** (filling missing values with the average of available data for that country).
 - *Why?* Ensures no gaps disrupt clustering.
- **Standardization:**
 - Applied StandardScaler to scale data (mean=0, variance=1).
 - *Why?* K-Means is sensitive to uneven scales; standardization ensures fairness.

2. Clustering with K-Means

Goal: Group countries with similar enrollment trends.

Key Concepts:

- **K-Means Clustering:**
 - An algorithm that partitions data into k groups ("clusters") based on similarity.
 - Countries in the same cluster have **similar enrollment patterns over time**.
- **Results (k=4 Clusters):**
 - **Cluster 0:** High gender parity (consistently near/above 1.0 ratio).

- **Cluster 1:** Steady improvement toward parity.
- **Cluster 2:** Persistent disparity (low ratios, slow progress).
- **Cluster 3:** Unstable trends (fluctuations/regressions).

Why This Matters:

Clusters reveal **global patterns** to guide policy (e.g., Cluster 2 needs urgent intervention).

3. Time-Series Visualization

Goal: Show real-world examples of trends per cluster.

- Plotted enrollment ratios over time for **one representative country per cluster**.
- *Example:* A country in Cluster 1 shows gradual progress from disparity (ratio < 1) to parity (ratio $\approx$ 1).

4. Dimensionality Reduction (PCA & t-SNE)

Goal: Visualize high-dimensional data in 2D for better interpretation.

Key Techniques:

1. **PCA (Principal Component Analysis):**

- Compresses data into 2 axes ("principal components") capturing ~70–80% of variance.
- *Result:* Countries from the same cluster grouped together in 2D space.

2. **t-SNE (t-Distributed Stochastic Neighbor Embedding):**

- Focuses on preserving **local similarities** (nonlinear patterns).
- *Result:* Tighter, more separated clusters than PCA.

Why Both?

- **PCA:** Fast, linear overview.
 - **t-SNE:** Reveals nuanced, nonlinear relationships.
-

5. Insights & Policy Implications

Cluster	Trend	Policy Action
0 (High parity)	Stable ratios	Sustain policies
1 (Improving)	Steady growth	Continue support
2 (Disparity)	Low ratios	Urgent intervention
3 (Irregular)	Fluctuations	Investigate causes

6. Conclusion

- **K-Means** grouped countries by enrollment trends.
- **PCA/t-SNE** confirmed clusters were meaningful.
- **Actionable Insight:** Clusters help target policies (e.g., prioritize aid to Cluster 2).

Key Tools Used:

- Python libraries: scikit-learn (K-Means, PCA), mlxtend (t-SNE), pandas (data wrangling).

This workflow demonstrates how **data mining extracts insights** from a data warehouse (Project 1).